

AG-512B Gearbox Oil Analysis

Laboratory report OAL-AG512B-2026-0624 | Gearbox GBX-512B

1. Lubricant identity and local requirements

Current lubricant finding OIL-F-01: SynLube PAG ISO VG 320. The removed gearbox used Mineral EP ISO VG 220; that grade is historical context only.

parameter	current local requirement	operator basis
Viscosity at 40 C	288-352 cSt inclusive	Range
Water content	450 ppm maximum	<=
Iron concentration	45 ppm maximum	<=
Copper concentration	12 ppm maximum	<=

2. Results and sample validity

sample_id	sample date	validity	controlling use	viscosity 40 C	water	iron	copper	note
OIL-2720	2026-06-20	INVALID	Exclude from acceptance	310.2 cSt	880 ppm	72 ppm	14 ppm	Broken tamper seal; field blank elevated
OIL-2721	2026-06-23	VALID	Primary current sample	318.6 cSt	540 ppm	58 ppm	9 ppm	Current accepted sample
OIL-2722	2026-06-23	VALID	Confirmatory current sample	320.1 cSt	525 ppm	55 ppm	10 ppm	Independent bottle and chain of custody
OIL-2507	2025-02-14	VALID	SUPERSEDED historical sample	218.0 cSt	0.09%	49 ppm	7 ppm	Predates gearbox replacement and lubricant change

3. Interpretation notes

OIL-2720 is invalid and cannot support any passing or failing analyte result. OIL-2721 is the primary current sample, with OIL-2722 as confirmatory current evidence. Historical OIL-2507 water content converts from 0.09% to 900 ppm using $1\% = 10,000 \text{ ppm}$, but the sample remains superseded.